

Navigating the New Relic One interface

Once your Spring Boot application is running alongside the New Relic agent, you can monitor the flow of information being sent to the New Relic server. You can verify the correct configuration either by examining the logs in the designated logs folder or by accessing the New Relic One website to check for the presence of your application.

If the New Relic agent is successfully transmitting information to the server, log in to the New Relic website and navigate to the 'APM' and 'Services' section. In Figure 4, we can find the name of our application listed.

Within this dashboard, you can access various metrics such as throughput, latency, and transaction time. Additionally, the platform provides insights into the time taken by each database operation. You can also analyse the time taken by each transaction and the duration of each operation within the transaction.

By leveraging these features, you can gain comprehensive insights into the performance of your application and its underlying components, facilitating effective optimisation and troubleshooting.

Advanced features and functionality of NRQL

The New Relic Query Language (NRQL) serves as a powerful tool for querying and analysing data within New Relic Insights. NRQL enables users to access and scrutinise data collected from diverse sources, including New Relic's agents, to gain insights into their infrastructure, applications, and user experience. By utilising NRQL queries, users can visually analyse their data within New Relic Insights, creating charts and visualisations to aid in understanding and decision-making. Notably, NRQL's syntax closely resembles that of SQL, with minor variations in writing conventions.

Key NRQL keywords include:

- **SELECT:** The 'SELECT' keyword is utilised to specify a particular metric or a function applied to a metric.
- **FROM:** The 'FROM' keyword specifies the event or metric data source from which data is to be retrieved. It is fundamental in NRQL queries and is paired with the SELECT statement to define the data to be queried.
- **FACET:** The 'FACET' keyword segments data for analysis, akin to the 'GROUP BY' clause in SQL.
- **WHERE:** The 'WHERE' keyword filters data based on specified criteria, such as selecting data from a particular application.
- **TIMEWINDOW function:** This function specifies a time window during which data is captured, using syntax like 'SINCE 1 hour ago' and 'UNTIL' to specify the end time for data extraction.
- **COMPARE WITH:** The 'COMPARE WITH' keyword facilitates comparison of metrics between two different time ranges, aiding in trend analysis.

- **ORDER BY:** The 'ORDER BY' keyword arranges query results based on specified criteria, such as the number of transactions, in descending order.
- **Mathematical aggregation functions:** NRQL supports mathematical aggregation functions like *min()*, *max()*, *sum()*, etc, enabling further analysis and computation on metrics. By leveraging these advanced features and functionality of NRQL, users can gain deeper insights into their data, facilitating informed decision-making and optimisation efforts.

As an example, instructions for an NRQL query to retrieve the number of successful web transactions in the last 24 hours are provided below:

```
SELECT count (*) AS ' Number of Transactions ' ,
average ( duration ) AS ' Average Duration ' ,
max ( duration ) AS ' Max Duration '
FROM Transaction
WHERE appName = ' My Application '
AND transactionType = ' Web '
AND httpResponseCode = ' 200 '
FACET countryCode
SINCE 1 day ago
UNTIL 1 hour ago
LIMIT 10
ORDER BY ' Number of Transactions ' DESC
```

Configuration of alerts and notifications

Setting up alerts for specific metrics in New Relic involves a straightforward process. Follow these steps to configure alerts and receive notifications when certain metrics reach predefined threshold values.

Login and access alerts and AI section: Log in to New Relic and navigate to the 'Alerts and AI' section. Proceed to the 'Alert conditions' subsection.

Initiate new alert condition: Click on the '+ New Alert Condition' button located at the top right corner of the web page. This action opens a dialog section for configuring the alert condition.

Guided mode selection: Opt for the 'Use guided mode' option to simplify the alert setup process.

Select service and entity: Choose the 'Services – APM' option and proceed to the next step.

Specify website for alert: Select the website or application entity for which you want to create the alert. For example, we chose 'Spandan Website'.

Choose metric: Select the metric for which you want to set the alert, such as 'Response Time' under the 'Golden Metrics' section. Alternatively, create a new metric using the 'Other Metrics' option.

Set thresholds: Define thresholds for the metric to trigger the alert. Configure limits and intervals accordingly.

Aggregation time and interval: Specify the aggregation